3.2 AM–GM, Backwards

WORKSHEET

The question

STEP II 1993, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s2-q8

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

? **Ask yourself** Why might it be easier to go from 2^m numbers to 2^{m+1} numbers than from n to $n+1$?

STAGE 2 Doubling

? **Ask yourself** Split the 2^{m+1} numbers into a first half and a second half, and let P and Q be the arithmetic means of the two halves. What does the inductive hypothesis say about each half?

STAGE 3 Filling the gaps

? **Ask yourself** What is the average of the padded list $b_1, \ldots, b_{2^m}$?

Pause and try Finish the proof.

What's really going on

Cauchy's induction

Cauchy published this proof in his *Cours d'analyse* (1821), the book that began the rigorous treatment of analysis we discussed in Chapter 2. The technique is now called *forward–backward induction*: prove a statement for an infinite sequence of cases (here the powers of 2), and then show that the case n implies the case $n-1$. Together, these reach every n. Our Stage 3 jumps straight down from 2^m to n, but you can also do it one step at a time: padding $n-1$ numbers with one copy of their average shows that the case n implies the case $n-1$.

Many proofs of one inequality

AM–GM has dozens of proofs, and comparing them is a good way to see different styles of mathematics.

- **By convexity.** The logarithm is concave, so the average of the logarithms is at most the logarithm of the average. That is AM–GM after taking exponentials, and it is the subject of Section 3.3.
- **George Pólya's proof.** Use $e^{x-1} \geq x$ (the tangent line to e^{x-1} at $x=1$ lies below the curve). With A the mean, apply it to each a_i/A and multiply: $\prod \frac{a_i}{A} \leq e^{\sum(a_i/A-1)} = e^0 = 1$. Pólya said this proof came to him in a dream.
- **By smoothing.** If two of the numbers are unequal, replace them both by their average. The arithmetic mean is unchanged and, by the two-number case, the product increases. Repeating pushes all the numbers towards the average.

Optimisation without calculus

AM–GM turns many optimisation problems into one-liners. Of all boxes with a given surface area, which has the largest volume? Of all rectangles with a given perimeter, which has the largest area? (A square, by the two-number case: $xy \leq \left(\frac{x+y}{2}\right)^2$, with

equality when $x = y$.) The equality condition tells you where the optimum is, and the inequality tells you it really is the optimum, not just a stationary point. Compare the warnings of Section 3.1.

Try this next

1 **One step back.** Show directly that if AM–GM holds for n numbers, it holds for $n-1$ numbers, by padding with one copy of the average.

2 **An increasing sequence.** Apply AM–GM to the $n+1$ numbers $1+\frac{1}{n}, \dots, 1+\frac{1}{n}, 1$ (with n copies of $1+\frac{1}{n}$) to show that $\left(1+\frac{1}{n}\right)^n < \left(1+\frac{1}{n+1}\right)^{n+1}$.

3 **The best box.** A closed rectangular box has surface area S. Use AM–GM on the areas of three different faces to show that its volume is at most $(S/6)^{3/2}$, with equality for a cube.

4 **Harmonic mean.** By applying AM–GM to the numbers $1/a_i$, show that the harmonic mean $\dfrac{n}{1/a_1 + \cdots + 1/a_n}$ is at most the geometric mean.

Further reading

- **J. Michael Steele, *The Cauchy–Schwarz Master Class*** (Cambridge University Press). A wonderful, problem-driven book on inequalities, starting from exactly this level. Chapter 2 is about AM–GM.
- **Hardy, Littlewood & Pólya, *Inequalities*** (Cambridge University Press). The classic reference, written by three of the great analysts of the twentieth century.
- **Arthur Engel, *Problem-Solving Strategies*** (Springer). Chapter 7 is a large collection of inequality problems from olympiads.

The full guided solution is in Section 3.2:
https://beyondstep.keithmcnulty.org/ch03/am-gm-backwards.html